Zhuo Chen
Chalmers University of Technology, SE-412 96 Gothenburg, Sweden
Email: zhuoc@chalmers.se

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Editors-in-Chief
Smart Agricultural Technology

Subject: Submission of a Research Article
Bounded Instance-Support Evaluation for Finite Crop–Weed Review Queues

Dear Editors-in-Chief,

On behalf of my co-authors, I am pleased to submit the manuscript, "Bounded Instance-Support Evaluation for Finite Crop–Weed Review Queues," for consideration as a Research Article in *Smart Agricultural Technology.*

Agricultural segmentation systems are commonly summarised by image- or instance-level averages, whereas practical quality assurance often operates through a finite queue of predicted regions. The manuscript introduces bounded instance-support evaluation, an executable contract that follows distinct plants through candidate formation, agricultural-role qualification, ranking, and capacity truncation. It separates spatial support, role-qualified support, merge-aware set coverage, and one-to-one support, thereby linking segmentation output to the number of plant records that a fixed review queue can represent.

On the official WeedsGalore split, proposal commissioning increased spatial support from 0.2973 to 0.5713 and role-qualified support from 0.1928 to 0.3026. Across 27 matching contracts and four ranker-training schemes, the queued effect depended on the proposal distribution used to train the ranker. A matched comparison of U-Net, DeepLabV3+, and Mask R-CNN at $K = 20$ placed Mask R-CNN highest at 0.2159 one-to-one support; its crossed seed-by-image contrast with U-Net was 0.0132 (95% interval, -0.0009 to 0.0300). Merge-aware evaluation additionally exposed a 105-instance difference between set coverage and one-to-one support.

The technical endpoint is connected to human review and transfer behaviour. Two independent operators assessed the same 518-region queue and reached 0.9421 agreement on weed reviewability (Cohen's $\kappa = 0.8930$). A cross-dataset study applied five validation-selected U-Net checkpoints to 60 Crop/Weed Field Image Dataset images and showed how score-scale and agricultural-role qualification jointly shape external queue formation.

The work aligns with *Smart Agricultural Technology* by integrating crop–weed segmentation, finite-capacity human review, matching design, and external calibration into a reproducible agricultural quality-assurance workflow. The accompanying public repository contains the configurations, source data for figures and tables, per-image and per-seed outputs, matching records, hashes, and replay programs needed to inspect the reported results.

The manuscript is original, has not been published, is not under consideration elsewhere, and has been approved by all authors. The authors declare no known competing interests.

Thank you for considering this manuscript. We believe it will interest readers working on agricultural vision, crop–weed management, model evaluation, and human-centred quality assurance.

Sincerely,

Zhuo Chen
on behalf of all authors